# SE YOON LEE

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## EDUCATION

**Ph.D. Statistics, Texas A&M University, College Station, U.S.A.** *May 2021*
Advisor: Prof. Bani K. Mallick Overall GPA: 3.78/4

**M.A. Applied Statistics, Yonsei University, Seoul, South Korea** *Feb 2016*
Advisor: Prof. Joseph H.T. Kim Overall GPA: 3.86/4

**B.S. Mathematics, Yonsei University, Seoul, South Korea** *Feb 2013*
Graduated with the 2nd highest class rank out of 43 students Overall GPA: 3.84/4

## RESEARCH INTERESTS

Causal Inference, Semiparametric Efficiency Theory, Bayesian Analysis, Clinical Trials, Biostatistics, ML/AI

## DISSERTATION AND THESIS

- "Bayesian Hierarchical Modeling: Application towards Complex and High-dimensional Data," *Texas A&M University*, Doctoral Dissertation ; **[Link]**
- "Exponentiated generalized Pareto distribution: an alternative to the generalized Pareto distribution," *Yonsei University*, Master's Thesis ; **[Link]**

## PROFESSIONAL EXPERIENCE

**AbbVie Inc.** *Feb 2025 – present*
***Manager, Statistics (Permanent Employment)*** North Chicago, IL

- Conducted research on modern causal inference methods for augmenting clinical trials with real-world data (RWD).
- Developed advanced survival analysis methods in causal–RWD settings.
- Supported and led various studies within the RWD group (e.g., observational studies, Health Technology Assessment submissions).
- Actively established myself as a causal inference expert (grounded in semiparametric efficiency theory and ML/AI) with strong domain knowledge in the pharmaceutical industry and academia by participating in conferences and publishing in renowned journals.

**Edwards Lifesciences.** *May 2023 – Feb 2025*
***Manager, Biostatistics (Permanent Employment)*** Irvine, CA

- Developed an R code to perform unblinded sample size re-estimation (promising zone approach) using CHW test statistics for a hierarchical composite endpoint, employing the Finkelstein and Schoenfeld methodology.
- Specialized in developing simulation R code for power calculation of Bayesian adaptive design for cardiovascular trials at the regulatory submission level, with expertise in end-to-end coding and Statistical Analysis Plan writing.
- Provided seminars on the topic of Bayesian Medical Device Trials to train biostatisticians ; **[Link]**
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to cross-functional teams.

**Johnson & Johnson.** *Apr 2022 – May 2023*
***Senior Biostatistician (Permanent Employment)*** Irvine, CA

- Developed Bayesian adaptive clinical trials designs to evaluate the safety and effectiveness of medical devices (Bayesian basket trial designs, power-prior designs, Bayesian group sequential designs, etc).
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to clinicians with main focus on cardiovascular diseases.
- Led working groups of experts (Machine Learning/Statistical Methodology & Consulting) as leader
- Led the development of basket trial designs and authored Statistical Analysis Plan for SECURE, a Post-Market Study [ClinicalTrials.gov ID: NCT04750798]

**Amgen Inc.** *May 2021 – Apr 2022*
***Scientist - Modeling & Simulations (Permanent Employment)*** Thousand Oaks, CA

- Developed Pharmacokinetics/Pharmacodynamics/Cox hazard regression models for Phase I cancer clinical trials of AMG160 to treat a metastatic castration-resistant prostate cancer (mCRPC).
- Researched machine learning and deep neural network models to fit single dose data for subsequent simulation of multiple dosing scenarios.

**Novartis International AG.** *May 2020 – Aug 2020*
***Biostatistics Summer Intern (Contract Employment)*** East Hanover, NJ

- Developed a Bayesian linear mixed effect model for patients who have wet age-related macular degeneration to predict best-corrected visual acuity over the maintenance phase and suggest a personalized dose regimen. The proposed model has been trained by actual patients' data from HAWK and HARRIER studies (Number of patients is around 1,800 patients). ; **[Abstract]**

**EMD Serono Inc. Merck KGaA.** *May 2019 – Aug 2019*
***Pharmacometrics Summer Intern (Contract Employment)*** Billerica, MA

- Developed a Bayesian adaptive clinical trial design in Phase I cancer clinical trials, which aimed at utilizing grade information from the Common Toxicity Criteria for Adverse Events provided by the National Cancer Institute. ; **[Abstract]** ; **[Poster]**

**Texas A&M University.** *Aug 2016 – May 2021*
***Graduate Research/Teaching Assistant*** College Station, TX

- Researched on applications and developments of various statistical models (e.g. non-linear mixed effect model, nonparametric/semiparametric longitudinal model, clustering analysis, classification, hierarchical Poisson model, etc) to various industrial problems arising from biomedical, petroleum, wind energy industries, and COVID-19 outbreak.
- Served as a Teaching Assistant for both undergraduate and graduate courses.

## PUBLICATIONS

### Peer-Reviewed Publications (*: corresponding author)

[1] **Se Yoon Lee*** and Jae Kwang Kim. (2026) "MEC: Machine-Learning-Assisted Generalized Entropy Calibration for Semi-Supervised Mean Estimation," *Forty-third International Conference on Machine Learning* (Accepted; 6,352 accepted among 23,918 reviewed submissions; acceptance rate: 26.6%) ; **[Video Recording]**; **[OpenReview]**

[2] **Se Yoon Lee***. (2026) "Asymptotic validity of Schoenfeld's sample size formula for the Cox proportional hazards model via the Wald test approach," *Statistical Methods in Medical Research*, 35(4), 681–694.

[3] **Se Yoon Lee***. (2025) "Power priors and type I error control: constrained borrowing of external control data," *Journal of Biopharmaceutical Statistics*

[4] **Se Yoon Lee***. (2025) "A note on the sample size formula for a win ratio endpoint," *Statistics in Medicine*, 44(15–17), e70165.

[5] **Se Yoon Lee***, Peng Zhao, Debdeep Pati, Bani K. Mallick. (2024) "Tail-adaptive Bayesian Shrinkage," *Electronic Journal of Statistics*

[6] **Se Yoon Lee***. (2024) "Eliciting the discount parameter in a power prior method on the basis of the type I error consideration," *Statistics in Biopharmaceutical Research*

[7] **Se Yoon Lee***. (2024) "Using Bayesian Statistics in Confirmatory Clinical Trials in the Regulatory Setting: A Tutorial Review," *BMC Medical Research Methodology*

[8] **Se Yoon Lee***. (2023) "A flexible dose-response modeling framework based on continuous toxicity outcomes in phase I cancer clinical trials," *BMC Trials*

[9] Abhinav Prakash, **Se Yoon Lee**, Xin Liu, Lei Liu, Bani Mallick, Yu Ding*. (2023) "A Bayesian hierarchical model to understand the effect of terrain on wind turbine power curves," *IEEE Transactions on Sustainable Energy*

[10] **Se Yoon Lee***. (2022) "The Use of a Log-Normal Prior for the Student t-Distribution," *Axioms*

[11] **Se Yoon Lee***. (2022) "Bayesian Nonlinear Models for Repeated Measurement Data: An Overview, Implementation, and Applications," *Mathematics*

[12] **Se Yoon Lee**, Alain Munafo, Pascal Girard, and Kosalaram Goteti*. (2022) "Optimization of dose selection using multiple surrogates of toxicity as a continuous variable in Phase I cancer trial," *Contemporary Clinical Trials*; **[Github]**

[13] **Se Yoon Lee***. (2021) "Gibbs sampler and coordinate ascent variational inference: a set-theoretical review," *Communications in Statistics - Theory and Methods*

[14] **Se Yoon Lee*** and Bani K. Mallick. (2021) "Bayesian Hierarchical modeling: application towards production results in the Eagle Ford Shale of South Texas," *Sankhyā: The Indian Journal of Statistics, Series B*; **[Github]**

[15] Kahkashan Afrin⋆, Ashif Iquebal⋆, Mostafa Karimi⋆, Allyson Larsen⋆, **Se Yoon Lee**⋆, and Bani K. Mallick⋆,*. (2020) "Directionally Dependent Multi-View Clustering Using Copula Model," *PLOS ONE* (⋆: equal contribution; authors are alphabetically ordered by last name.)

[16] **Se Yoon Lee***, Bowen Lei, and Bani K. Mallick. (2020) "Estimation of COVID-19 spread curves integrating global data and borrowing information," *PLOS ONE*; **[Github]**

[17] **Seyoon Lee**, Joseph H.T. Kim*. (2018) "Exponentiated generalized Pareto distribution: Properties and applications towards extreme value theory," *Communications in Statistics - Theory and Methods*, 48:8, 2014–2038

## UNDER REVIEW

[1] **Se Yoon Lee***. (2026+) "Reliable Causal Inference for Externally Controlled Single-Arm Trials under Transportability Violations, " *Under Review (contact me for the preprint)*

[2] **Se Yoon Lee***, Jieying Jiang, David Chen, Weili He, Dai Feng. (2026+) "Applying Ensemble Survival Models to Predict Median Survival Time in Health Technology Assessment, " *Under Review*

[3] **Se Yoon Lee***, Yonghyun Kwon, Jae Kwang Kim. (2026+) "MEC-Cox: Machine-Learning-Assisted Generalized Entropy Calibration for ATT Marginal Hazard-Ratio Estimation, " *Under Review*

## PROFESSIONAL ACTIVITIES

### Editorial Board Member

· BMC Medical Research Methodology (Impact Factor: 3.9)

### Invited Peer Reviewers

· Neural Information Processing Systems (h5-index : 371)
· International Conference on Machine Learning (h5-index : 272; Chosen as a Gold Reviewer)
· Artificial Intelligence Review (Impact Factor : 8.139)
· Nature Medicine (Impact Factor : 58.7)
· Statistics in Medicine (Impact Factor: 2.373)
· Biostatistics (Impact Factor: 1.8)

- Pharmaceutical Statistics (Impact Factor: 1.77)
- Statistical Methods in Medical Research (Impact Factor: 1.6)
- Bayesian Analysis (Impact Factor: 3.728)
- Contemporary Clinical Trials Communications (Impact Factor: 1.4)
- European Journal of Clinical Investigation (Impact Factor: 3.481)
- The Journal of Clinical Pharmacology (Impact Factor: 3.126)
- Frontiers in Public Health (Impact Factor: 3.709)
- The R Journal (Impact Factor: 3.984)
- Computational Geosciences (Impact Factor: 2.948)
- Journal of Applied Statistics (Impact Factor: 1.416)

## HONORS AND AWARDS

**Springer Nature Editor of Distinction Awards 2026** *May 2026*
Recognized among the top 20% of editors for editorial service to BMC Medical Research Methodology

**Travel fund for an invited talk at Yonsei University** *Dec 2019*
Given to an invited speaker for the presentation

**Travel fund for an invited talk at University of Michigan** *Jun 2019*
Given to an invited speaker for an annual meeting about wind energy

**Travel fund for poster presenter for the Houston Geological Society** *Mar 2018*
Given to a poster presenter

**Anant Kshirsagar fellowship** *Jul 2018*
Given to graduate students who demonstrate excellence in making progress in research

**The 1st place prize in SETCASA poster session in 2018** *Apr 2018*
Given to only one winner for the poster session

**Graduate student travel award for JSM** *Jul 2017, 2018*
Given to graduate students who make a presentation at the conference

## SOFTWARE

**Software used in papers**

- Bayesian Hierarchical Richards Model ; Written in R ; **[Download]** ; **[Github]**
- Spatial Weibull Model ; Written in R ; **[Github]**
- winratiosim ; Written in R ; **[Github]**

## SKILLS

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| **Language** | Fluent in English and Korean; Elementary in Chinese and Japanese |
| **Computer Language** | Proficient in R, Python, SQL, Microsoft Access, SAS, and NONMEM |
| **Certificate** | Cognigen NONMEM Workshop |

## REFERENCES

| | |
|---|---|
| **Bani K. Mallick** | Distinguished Professor, Texas A&M University, bmallick@stat.tamu.edu |
| **Debdeep Pati** | Professor, University of Wisconsin - Madison, dpati2@wisc.edu |
| **Samiran Sinha** | Professor, Texas A&M University, sinha@stat.tamu.edu |
| **Alan Dabney** | Professor, Texas A&M University, adabney@stat.tamu.edu |